

The empirical recurrence for OEIS A232672: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

7 September 2026

Abstract

OEIS A232672 records “Empirical recurrence of order 53”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 87$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 53, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 53 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A232672 is “Number of $(n+1) \times (3+1)$ 0..2 arrays with every element next to itself plus and minus one within the range 0..2 horizontally, vertically, diagonally or antidiagonally.”. It has offset 1 and begins

1568, 63632, 3684840, 223396744, 12890094328, 754650484792, 44225213151124, 2584211741776292, ...

The entry states:

Empirical recurrence of order 53 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 07:33:30, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 6642$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 87$ of the 6642, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 87$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 13284$ exact terms of the model returns order 53 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{53} c_i a(n-i),$$

c_1	23	c_{15}	1202933246698358	c_{29}	749713986925877274496	c_{43}	34894768305813
c_2	1179	c_{16}	3823724345194900	c_{30}	−312060840208458881920	c_{44}	−35585300808553
c_3	47801	c_{17}	22815509832425644	c_{31}	857203039925459213056	c_{45}	−1009878366807
c_4	295142	c_{18}	−63891636229773392	c_{32}	−2118610568859290996480	c_{46}	1177579876408
c_5	349830	c_{19}	−137632450945124168	c_{33}	764020647115734063104	c_{47}	3976387999967
c_6	−83258146	c_{20}	−854077186917743224	c_{34}	−2343918360887695571968	c_{48}	1728472991288
c_7	−579202036	c_{21}	1337960723399414896	c_{35}	3461303669449446682624	c_{49}	−774752977396
c_8	−2890497677	c_{22}	1730043927112435920	c_{36}	−997113903590955950080	c_{50}	−182178383971
c_9	44312348515	c_{23}	15236440613333828832	c_{37}	2877387424387989254144	c_{51}	−73282889795
c_{10}	284463556971	c_{24}	−14043938165667360992	c_{38}	−3193367094809222885376	c_{52}	513478527241
c_{11}	1319519035341	c_{25}	−2541855592070908928	c_{39}	569451319039215796224	c_{53}	−24030926136
c_{12}	−10268358688788	c_{26}	−144888329201570062752	c_{40}	−1635231012910299742208		
c_{13}	−50590492259292	c_{27}	82700419848167842080	c_{41}	1540001104831457918976		
c_{14}	−260280305470468	c_{28}	−111910318727832577600	c_{42}	1046559358238326784		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 53, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $53 + 4$ terms removed returns order 53 as well, so $\deg q_{\min} = 53 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $53 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 53 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A232672.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.